ENDG 319 - Fall 26
CURE Project – Deliverable 2

Graphical data analysis to develop a machine-learning model

Instructions:

- This deliverable is worth 3% of your overall grade and should be submitted in a group.
- The assignment requires you to use Python. Paste screenshots of your code and outputs into a Word file to show your work. Please keep the original Word file. Once done, convert the Word file into PDF with the file name: ENDG319_group number_CURE Deliverable 2.
- Submit the PDF file to Assessments > Dropbox > 'CURE Project Deliverable 2' by **October 18, 2026, 11:59 pm (MT)**.
- You must submit this deliverable on time to be able to submit the upcoming deliverables.

Research skills:

(i) Dealing with real-world data
(ii) Graphical analysis of data, interpreting figures
(iii) Critical thinking – drawing conclusions from data
(iv) Use of engineering tools – Python
(v) Developing a research question

Relevant course content: Descriptive Statistics (Ch. 1), Machine learning – classification

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Exploring the breast_cancer dataset in scikit-learn

The breast_cancer dataset contains 569 instances, 30 features and 2 classes, as shown below.

```
In [1]: import sklearn.datasets
        cancer = sklearn.datasets.load_breast_cancer()

In [2]: cancer.keys()
Out[2]: dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename'])

In [3]: print(cancer.feature_names)
['mean radius' 'mean texture' 'mean perimeter' 'mean area'
 'mean smoothness' 'mean compactness' 'mean concavity'
 'mean concave points' 'mean symmetry' 'mean fractal dimension'
 'radius error' 'texture error' 'perimeter error' 'area error'
 'smoothness error' 'compactness error' 'concavity error'
 'concave points error' 'symmetry error' 'fractal dimension error'
 'worst radius' 'worst texture' 'worst perimeter' 'worst area'
 'worst smoothness' 'worst compactness' 'worst concavity'
 'worst concave points' 'worst symmetry' 'worst fractal dimension']

In [4]: print(cancer.target_names)
['malignant' 'benign']
```

More information is available in the dataset description. Read the following snippet from the description.

```
In [5]: print(cancer.DESCR)
    :Class Distribution: 212 - Malignant, 357 - Benign

    :Creator:  Dr. William H. Wolberg, W. Nick Street, Olvi L. Mangasarian

    :Donor: Nick Street

    :Date: November, 1995

This is a copy of UCI ML Breast Cancer Wisconsin (Diagnostic) datasets.
https://goo.gl/U2Uwz2

Features are computed from a digitized image of a fine needle
aspirate (FNA) of a breast mass.  They describe
characteristics of the cell nuclei present in the image.
```

Note: the short link printed in the description (goo.gl) is no longer active. The original dataset page is in the UCI Machine Learning Repository: Breast Cancer Wisconsin (Diagnostic), https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

Task 1

Create a dataframe variable ‘a’ with this dataset. This dataframe should have all 569 instances, the 30 features, and the class of each instance as 0 (Malignant) or 1 (Benign). The column that contains the classes should be labeled ‘typeofcancer’.

(a) Show the output of a.shape.

```
[23]: a.shape
```

[Hints: the outputs should be same as below.]

```
[23]: (569, 31)
```

(b) Now create a dataframe variable ‘df’ by slicing dataframe ‘a’. The new dataframe ‘df’ should have all the instances and their labels, but only the following three features: mean radius, mean perimeter and mean area, along with the typeofcancer column. [Hint: use the .iloc method to extract the necessary columns from ‘a’.]

(i) Show the first two rows. [Hint: the output should be the same as below.]

[53]:

	mean radius	mean perimeter	mean area	typeofcancer
0	17.99	122.8	1001.0	0
1	20.57	132.9	1326.0	0

(ii) Show the rows with indexes 17, 18, 19, 20, 21. [Hint: the output should be the same as below.]

[54]:

	mean radius	mean perimeter	mean area	typeofcancer
17	16.130	108.10	798.8	0
18	19.810	130.00	1260.0	0
19	13.540	87.46	566.3	1
20	13.080	85.63	520.0	1
21	9.504	60.34	273.9	1

Task 2

Suppose we want to explore the possibility of developing a machine learning model that can diagnose a new patient's cancer condition as benign or malignant from the features in variable a.

(i) As a first step, you want to do some graphical analysis. Write the code to generate the following figure (Figure 1). Show screenshots of the code (input) and the figure (output) from your work. You are free to choose your favourite data marker and colour in your figure.

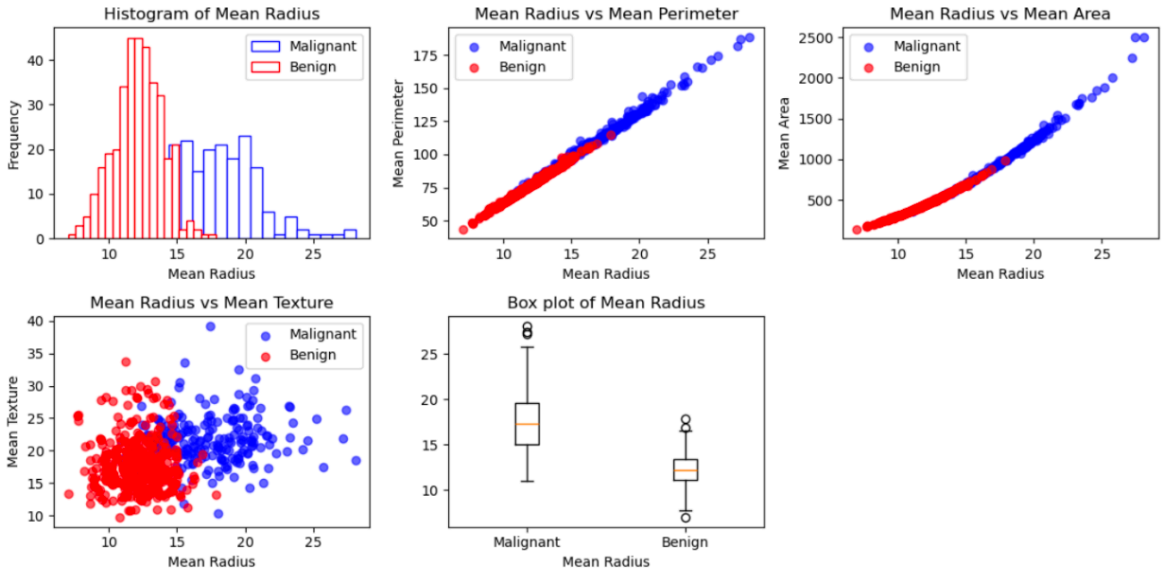


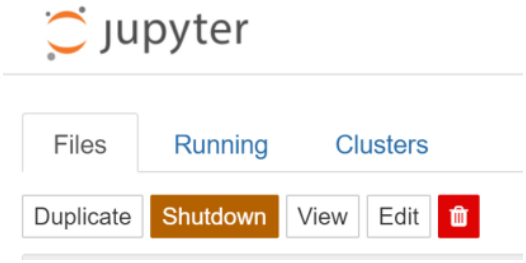
Figure 1: From left to right: histogram of 'mean radius' data for each class, scatter plot of 'mean radius' versus 'mean perimeter', scatter plot of 'mean radius' versus 'mean area', scatter plot of 'mean radius' versus 'mean texture', and box plot of 'mean radius' for each class.

(ii) Briefly describe what each of the first three subplots in Figure 1 reveals about the data.

Task 3

Save the file generated in Task 2. Make a copy of it and modify the code to generate the following figure (Figure 2).

Note: In Jupyter you can make a copy of a notebook with 'Duplicate'. In JupyterLab, right-click the notebook in the file browser and choose 'Duplicate'. In the classic Notebook interface, select the file and click the 'Duplicate' button, as shown below.



(i) Generate the following figure (Figure 2). Show screenshot(s) of the code (input) and the figure (output) from your work. You are free to choose your favourite data marker and colour in your figure.

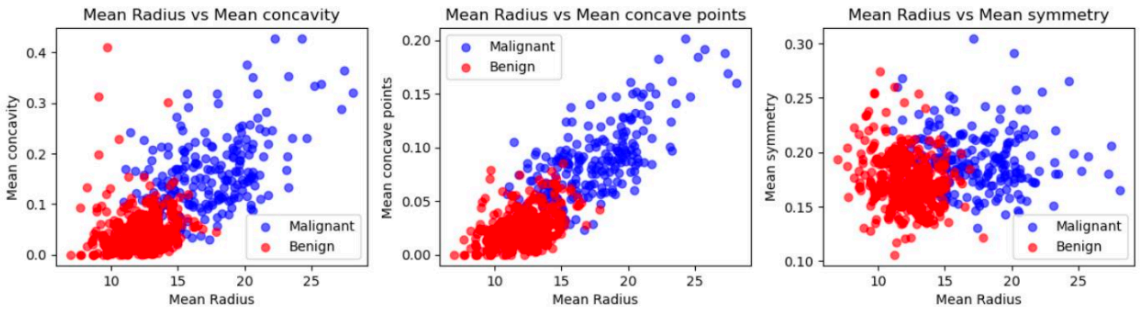


Figure 2: From left to right: scatter plot of 'mean radius' versus 'mean concavity', scatter plot of 'mean radius' versus 'mean concave points', and scatter plot of 'mean radius' versus 'mean symmetry'.

(ii) Briefly describe what each of the subplots in Figure 2 reveals about the data.

Task 4

Think of a classification problem from your own experience and interest (e.g., distinguishing between two types of things, living beings, weather, personalities, vehicles, industries, sports, houses, professionals, etc.). Collect some real-world data to solve that classification problem. Have at least two classes, two attributes and at least 5 instances per class. Create a dataframe with this data and paste a screenshot of it. Also give a short description of the data (number of instances, number of attributes and the list of attributes, list of classes, and creator: your name).

The classification problem should be a new one, but the dataset may contain data that are new (generated by you through rough measurements) or collected from the public domain. Please cite the references/sources from which you collected the data.